

Anand D. Sarwate

Curriculum Vitae

Professor

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EMPLOYMENT

- 1/14–present **Rutgers, The State University of New Jersey**, (Piscataway, New Jersey USA)
Professor (7/25–present), Associate Professor (7/20–6/25), Assistant Professor (1/14–6/20)
- 10/11–12/13 **Toyota Technological Institute at Chicago**, (Chicago, Illinois USA)
Research Assistant Professor
- 9/08–9/11 **University of California, San Diego**, (La Jolla, California USA)
Postdoctoral Researcher
Supervisors: Professors Alon Orlitsky, Tara Javidi, and Young-Han Kim

VISITING POSITIONS

- 7/26–6/27 Visiting Scholar, Department of Pure Mathematics and Mathematical Statistics, Cambridge University (Cambridge, UK)
- 1/15/19–5/17/19 Visiting Scholar, Simons Institute for Theoretical Computer Science (Berkeley, CA)

EDUCATION

- 1/06–7/08 **University of California, Berkeley**, (Berkeley, California USA)
Ph.D., Electrical Engineering and Computer Sciences (awarded 12/2008)
Designated Emphasis in Communication, Computation and Statistics
Thesis: *Robust and adaptive communication under uncertain interference*
Advisor: Professor Michael Gastpar
- 8/02–12/05 **University of California, Berkeley**, (Berkeley, California USA)
M.S., Electrical Engineering and Computer Sciences (awarded 12/2005)
Thesis : *Observation uncertainty in Gaussian sensor networks*
Advisor: Professor Michael Gastpar
- 9/97–6/02 **Massachusetts Institute of Technology**, (Cambridge, Massachusetts USA)
B.S., Electrical Science and Engineering (awarded 6/2002)
B.S., Mathematics (awarded 6/2002)
Minor in Music
Minor in Theater Arts

AWARDS AND HONORS

- 2025 Presidential Outstanding Faculty Scholar Award, Rutgers University
- 2025 Rutgers ECE Department Impact Award
- 2024–2025 IEEE Information Theory Society Distinguished Lecturer
- 2023 Outstanding Engineering Professor, Rutgers School of Engineering
- 2020 Board of Trustees Research Fellowship for Scholarly Excellence, Rutgers University, 2020

2018 A. Walter Tyson Assistant Professor Award, Rutgers School of Engineering
 2015 NSF CAREER Award

RECENT JOURNAL AND ARCHIVAL CONFERENCE PAPERS

A full list can be found on my website.

- [1] Tao, Y. et al. Oct. 2026. “Privacy-Preserving Visualization of Brain Functional Connectivity”. In: *NeuroImage* 340.
- [1] Tao, Y. and Sarwate, A. D. Sept. 2026. “Functional Approximation Methods for Differentially Private Distribution Estimation”. In: *IEEE Transactions on Information Forensics and Security* 21, pp. 8241–8255.
- [1] Vargas, M., Cannon, R., Engel, A., Sarwate, A. D., and Chiang, T. Nov. 2025. “Understanding generative AI output with embedding models”. In: *Science Advances* 11.48.
- [1] Sarwate, A. D., Calmon, F. P., Kosut, O., and Sankar, L. Oct. 2025. “An Information Theorist’s Tour of Differential Privacy”. In: *IEEE BITS the Information Theory Magazine* 4.4, pp. 2–15.
- [1] Banerjee, S., Marrinan, T., Cannon, R., Chiang, T., and Sarwate, A. D. Oct. 2025. “Measuring training variability from stochastic optimization using robust nonparametric testing”. In: *IEEE Journal of Selected Topics in Signal Processing* 19.7, pp. 1477–1492.
- [1] Wu, Y., Li, Y., Dong, Z., Sathyavageswaran, N., and Sarwate, A. D. Apr. 2025. “Learning to Help in Multi-Class Settings”. In: *The Thirteenth International Conference on Learning Representations (ICLR 2025)*.
- [1] Engel, A. W., Wang, Z., Frank, N., Dumitriu, I., Choudhury, S., Sarwate, A., and Chiang, T. May 2024. “Faithful and Efficient Explanations for Neural Networks via Neural Tangent Kernel Surrogate Models”. In: *The Twelfth International Conference on Learning Representations*. Vienna, Austria.
- [1] Wang, Z., Engel, A., Sarwate, A., Dumitriu, I., and Chiang, T. Dec. 2023. “Spectral Evolution and Invariance in Linear-width Neural Networks”. In: *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. Curran Associates, Inc.
- [1] Taki, B., Sarwate, A. D., and Bajwa, W. U. Aug. 2023. “Structured Low-Rank Tensor Models for Logistic Regression”. In: *Transactions on Machine Learning Research*.
- [1] Gazula, H. et al. Nov. 2022. “Federated Analysis in COINSTAC Reveals Functional Network Connectivity and Spectral Links to Smoking and Alcohol Consumption in Nearly 2,000 Adolescent Brains”. In: *Neuroinformatics* 21, pp. 287–301.
- [1] Zhang, Y., Vatedka, S., Jaggi, S., and Sarwate, A. D. Aug. 2022. “Quadratically Constrained Myopic Adversarial Channels”. In: *IEEE Transactions on Information Theory* 68, pp. 4901–4948.
- [1] Xiong, S., Sarwate, A. D., and Mandayam, N. B. June 2022. “Network Traffic Shaping for Enhancing Privacy in IoT Systems”. In: *IEEE/ACM Transactions on Networking* 30.3, pp. 1162–1177.
- [1] Cangialosi, F., Agarwal, N., Arun, V., Jiang, J., Narayana, S., Sarwate, A., and Netravali, R. Apr. 2022. “Privid: Practical, Privacy-Preserving Video Analytics Queries”. In: *Proceedings of the 19th USENIX Symposium on Networked Systems Design and Implementation (NSDI ’22)*.
- [1] Imtiaz, H., Mohammadi, J., Silva, R., Baker, B., Plis, S. M., Sarwate, A. D., and Calhoun, V. D. Nov. 2021. “A Correlated Noise-Assisted Decentralized Differentially Private Estimation Protocol, and its Application to fMRI Source Separation”. In: *IEEE Transactions on Signal Processing* 69, pp. 6355–6370.
- [1] Kurri, G. R., Prabhakaran, V. M., and Sarwate, A. D. Aug. 2021. “Coordination Through Shared Randomness”. In: *IEEE Transactions on Information Theory* 67.8, pp. 4948–4974.
- [1] Nikolakakis, K. E., Kalogerias, D. S., Sarwate, A. D., and Sheffet, O. June 2021. “Quantile Multi-Armed Bandits: Optimal Best-Arm Identification and a Differentially Private Scheme”. In: *IEEE Journal on Selected Areas in Information Theory* 2.2, pp. 534–548.
- [1] Nikolakakis, K. E., Kalogerias, D. S., and Sarwate, A. D. Apr. 2021. “Predictive Learning on Hidden Tree-Structured Ising Models”. In: *Journal of Machine Learning Research* 22.59, pp. 1–82.
- [1] Gazula, H. et al. Jan. 2021. “Decentralized Multisite VBM Analysis During Adolescence Shows Structural Changes Linked to Age, Body Mass Index, and Smoking: a COINSTAC Analysis”. In: *Neuroinformatics*.

- [1] Ghassemi, M., Shakeri, Z., Sarwate, A. D., and Bajwa, W. U. Jan. 2020. “[Learning Mixtures of Separable Dictionaries for Tensor Data: Analysis and Algorithms](#)”. In: *IEEE Transactions on Signal Processing* 68.1, pp. 33–48.
- [1] Hazan, T., Orabona, F., Sarwate, A. D., Maji, S., and Jaakkola, T. Oct. 2019. “[High Dimensional Inference with Random Maximum A-Posteriori Perturbations](#)”. In: *IEEE Transactions on Information Theory* 65.10, pp. 6539–6560.

BOOK CHAPTERS AND MONOGRAPHS

- [1] Dey, B. K., Jaggi, S., Langberg, M., Sarwate, A. D., and Zhang, Y. Dec. 2024. “[Codes for Adversaries: Between Worst-Case and Average-Case Jamming](#)”. In: *Foundations and Trends in Communications and Information Theory* 21.3-4, pp. 300–588.
- [1] Sarwate, A. D. Oct. 2024. “[Machine learning with differential privacy](#)”. In: *Handbook of Sharing Confidential Data Differential Privacy, Secure Multiparty Computation, and Synthetic Data*. Ed. by J. Drechsler, D. Kifer, J. Reiter, and A. Slavkovic. Chapman & Hall/CRC Handbooks of Modern Statistical Methods. Boca Raton, FL, USA: CRC Press. Chap. 7, pp. 91–114.
- [1] Shakeri, Z., Sarwate, A. D., and Bajwa, W. U. Mar. 2021. “[Sample Complexity Bounds for Dictionary Learning from Vector- and Tensor-valued Data](#)”. In: *Information-Theoretic Methods in Data Science*. Ed. by M. Rodrigues and Y. C. Eldar. Cambridge, UK: Cambridge University Press. Chap. 5, pp. 134–162.

RECENT CONFERENCE PAPERS

- [1] Sathyavageswaran, N., Sarwate, A. D., Mandayam, N. B., and Yates, R. D. Nov. 2026. “Age and Stability Trade-offs in Remote Monitoring Systems”. In: *Proceedings of the 2026 IEEE Information Theory Workshop (ITW 2026)*.
- [1] Tao, Y., Baker, B. T., Wu, Y., Sarwate, A. D., Panta, S., Plis, S., and Calhoun, V. D. July 2026. “[Batch Effects In Brain Foundation Model Embeddings](#)”. In: *ICML 2026 Workshop on Structured Data for Health (non-archival)*.
- [1] Dey, B. K., Jaggi, S., Langberg, M., Sarwate, A. D., and Zhang, Y. June 2025. “[Sliding Window Adversarial Channels](#)”. In: *Proceedings of the 2025 IEEE International Symposium on Information Theory (ISIT)*.
- [1] Tao, Y. and Sarwate, A. D. Apr. 2025. “[Differentially Private Distribution Estimation Using Functional Approximation](#)”. In: *Proceedings of the 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2025)*.
- [1] Basodi, S., Romero, J., Panta, S., Martin, D., Plis, S., Sarwate, A. D., and Calhoun, V. D. Apr. 2025. “[Federated Analysis of Neuroimaging Data Using COINSTAC](#)”. In: *2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI)*, pp. 1–4.
- [1] Basodi, S., Romero, J., Panta, S., Martin, D., Plis, S., Sarwate, A. D., and Calhoun, V. D. Dec. 2024. “[Advanced machine learning in neuroimaging studies via federated learning](#)”. In: *2024 IEEE International Conference on Big Data (BigData)*, pp. 8623–8624.
- [1] Tao, Y., Sarwate, A. D., Panta, S., Plis, S., and Calhoun, V. D. Dec. 2024. “[Federated Privacy-Preserving Visualization: A Vision Paper](#)”. In: *2024 IEEE International Conference on Big Data (BigData)*, pp. 8035–8041.
- [1] Sathyavageswaran, N., Yates, R. D., Sarwate, A. D., and Mandayam, N. B. Nov. 2024. “[Timely Offloading in Mobile Edge Cloud Systems](#)”. In: *Proceedings of the 2024 IEEE Information Theory Workshop*. Shenzhen, China, pp. 133–138.
- [1] Dey, B. K., Jaggi, S., Langberg, M., Sarwate, A. D., and Zhang, Y. July 2024. “[Computationally Efficient Codes for Strongly Dobrushin-Stambler Nonsymmetrizable Oblivious AVCs](#)”. In: *Proceedings of the 2024 IEEE International Symposium on Information Theory (ISIT)*. Athens, Greece.
- [1] Tao, Y., Sarwate, A. D., Panta, S., Plis, S., and Calhoun, V. D. May 2024. “[Privacy-Preserving Visualization of Brain Functional Network Connectivity](#)”. In: *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, pp. 1–5.

- [1] Sanchez, J. H., Taki, B., Bajwa, W., and Sarwate, A. Apr. 2024. “**Federated Learning of Tensor Generalized Linear Models with Low Separation Rank**”. In: *Proceedings of the 2024 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2024)*, pp. 2136–2140.
- [1] Taki, B., Sarwate, A. D., and Bajwa, W. U. Mar. 2024. “**Low Separation Rank in Tensor Generalized Linear Models: an Asymptotic Analysis**”. In: *2024 58th Annual Conference on Information Sciences and Systems (CISS)*. Princeton, NJ, USA, pp. 1–6.
- [1] Saha, D. K., Calhoun, V., Kwon, S. M., Sarwate, A., Saha, R., and Plis, S. July 2023. “**Federated, Fast, and Private Visualization of Decentralized Data**”. In: *Federated Learning and Analytics in Practice: Algorithms, Systems, Applications, and Opportunities (FL-ICML 2023)*.
- [1] Wang, Z., Engel, A., Sarwate, A., Dumitriu, I., and Chiang, T. July 2023. “**Spectral Evolution and Invariance in Linear-width Neural Networks**”. In: *Workshop on High-dimensional Learning Dynamics (HiLD-ICML 2023)*.
- [1] Li, S., Krishan, P., Jaggi, S., Langberg, M., and Sarwate, A. D. June 2023. “**Computationally Efficient Codes for Adversarial Binary-Erasure Channels**”. In: *Proceedings of the 2023 IEEE International Symposium on Information Theory (ISIT)*, pp. 228–233.
- [1] Sathyavageswaran, N., Yates, R. D., Sarwate, A. D., and Mandayam, N. June 2022. “**Privacy Leakage in Discrete Time Updating Systems**”. In: *Proceedings of the 2022 IEEE International Symposium on Information Theory (ISIT)*.
- [1] Zhang, Y., Jaggi, S., Langberg, M., and Sarwate, A. D. June 2022. “**The Capacity of Causal Adversarial Channels**”. In: *Proceedings of the 2022 IEEE International Symposium on Information Theory (ISIT)*.
- [1] Kwon, S. M., Li, X., and Sarwate, A. D. May 2022. “**Low-Rank Phase Retrieval with Structured Tensor Models**”. In: *Proceedings of the 47th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2022)*, pp. 3643–3647.
- [1] Tao, Y., Chihoub, A., Sarwate, A. D., Panta, S., and Calhoun, V. 2022d. “Privacy-Preserving Visualization of Functional Network Connectivity”. In: *International Conference of the IEEE Engineering in Medicine and Biology Society*. Glasgow, Scotland, UK.
- [1] Taki, B., Ghassemi, M., Sarwate, A. D., and Bajwa, W. U. Nov. 2021. “**A Minimax Lower Bound for Low-Rank Matrix-Variate Logistic Regression**”. In: *2021 Asilomar Conference on Signals, Systems, and Computers*, pp. 477–484.
- [1] Rezaie, A., Gao, J., and Sarwate, A. D. Apr. 2021. “**Influencers and the Giant Component: the Fundamental Hardness in Privacy Protection for Socially Contagious Attributes**”. In: *SIAM International Conference on Data Mining*. Ed. by C. Demeniconi and I. Davidson, pp. 217–225.
- [1] Budkuley, A. J., Dey, B. K., Jaggi, S., Langberg, M., Sarwate, A. D., and Wang, C. June 2020. “**Symmetrizability for Myopic AVCs**”. In: *Proceedings of the 2020 IEEE International Symposium on Information Theory (ISIT)*.

EDITORIAL AND PROFESSIONAL SERVICE

10/21–ongoing	Associate Editor, IEEE Transactions on Information Theory
1/20–ongoing	Consulting Associate Editor, IEEE Open Journal of Signal Processing (OJSP)
2021–2026	Member, Board of Governors, IEEE Information Theory Society
2026	Tutorials Co-Chair, 2026 IEEE Information Theory Workshop (ITW 2026)
2024	Co-Organizer, DIMACS Workshop on Modeling Randomness in Neural Network Training
2017–2022	Member, Machine Learning for Signal Processing Technical Committee, IEEE Signal Processing Society